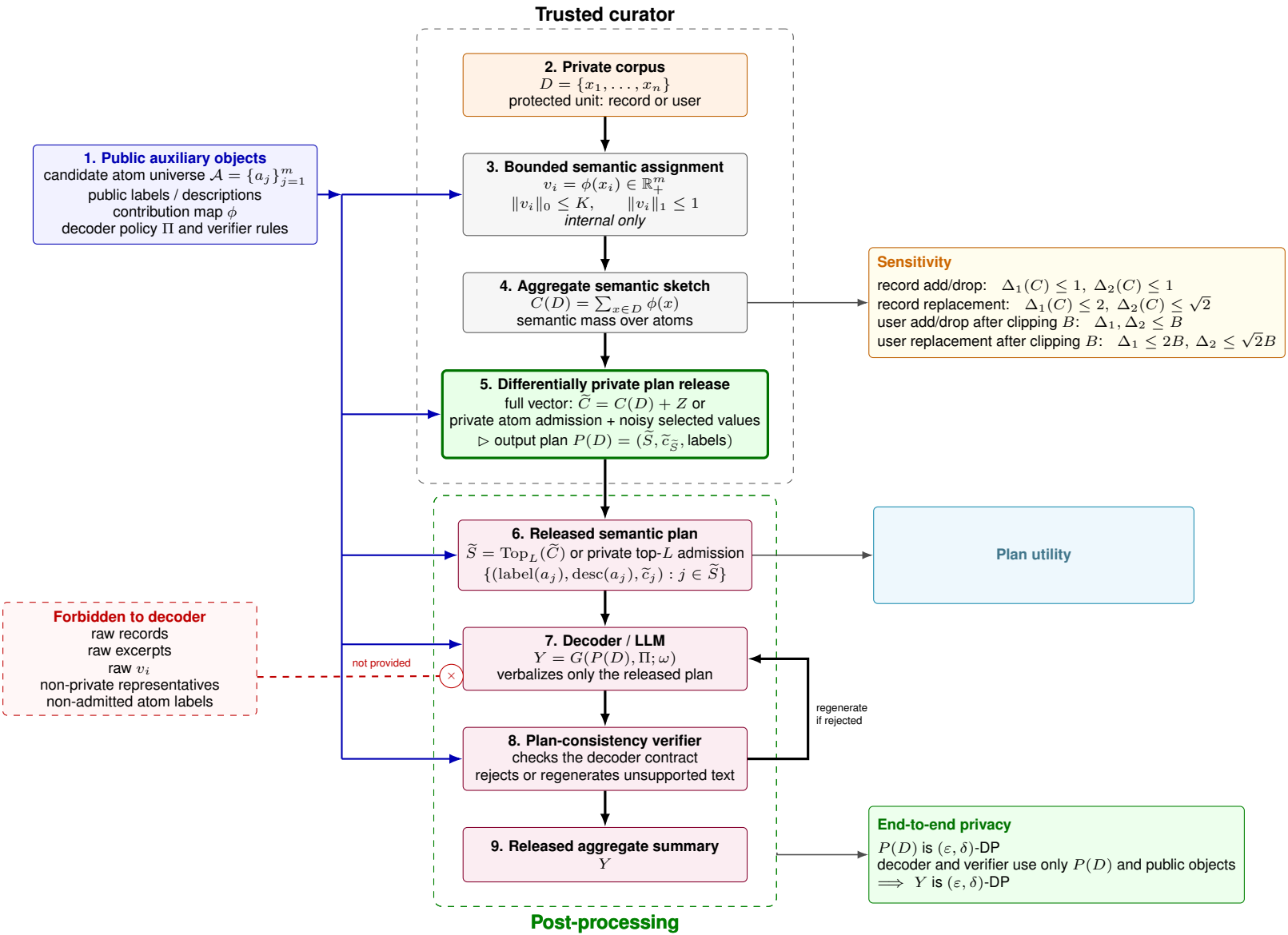

Trusted curator
2. Private corpus
$D = \{x_1, \ldots, x_n\}$
protected unit: record or user
1. Public auxiliary objects
candidate atom universe $\mathcal{A} = \{a_j\}_{j=1}^m$
public labels / descriptions
contribution map ϕ
decoder policy Π and verifier rules
3. Bounded semantic assignment
$v_i = \phi(x_i) \in \mathbb{R}_+^m$
$\|v_i\|_0 \le K, \quad \|v_i\|_1 \le 1$
internal only
4. Aggregate semantic sketch
$C(D) = \sum_{x \in D} \phi(x)$
semantic mass over atoms
Sensitivity
record add/drop: $\Delta_1(C) \le 1,\ \Delta_2(C) \le 1$
record replacement: $\Delta_1(C) \le 2,\ \Delta_2(C) \le \sqrt{2}$
user add/drop after clipping B: $\Delta_1, \Delta_2 \le B$
user replacement after clipping B: $\Delta_1 \le 2B,\ \Delta_2 \le \sqrt{2}B$
5. Differentially private plan release
full vector: $\widetilde{C} = C(D) + Z$ or
private atom admission + noisy selected values
▷ output plan $P(D) = (\widetilde{S}, \widetilde{c}_{\widetilde{S}}, \text{labels})$
6. Released semantic plan
$\widetilde{S} = \mathrm{Top}_L(\widetilde{C})$ or private top-L admission
$\{(\mathrm{label}(a_j), \mathrm{desc}(a_j), \widetilde{c}_j) : j \in \widetilde{S}\}$
Plan utility
Forbidden to decoder
raw records
raw excerpts
raw v_i
non-private representatives
non-admitted atom labels
not provided
7. Decoder / LLM
$Y = G(P(D), \Pi; \omega)$
verbalizes only the released plan
regenerate
if rejected
8. Plan-consistency verifier
checks the decoder contract
rejects or regenerates unsupported text
9. Released aggregate summary
Y
End-to-end privacy
$P(D)$ is (ε, δ)-DP
decoder and verifier use only $P(D)$ and public objects
$\Longrightarrow$ Y is (ε, δ)-DP
Post-processing